

BERKHOUT, Netherlands

WMO: 062490

Lat: 52.6428N

Lon: 4.9789E

Elev: -10

StdP: 14.70

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 21.0	(c) 24.6	(d) 14.8	(e) 11.6	(f) 23.1	(g) 19.6	(h) 14.7	(i) 25.4	(j) 36.3	(k) 48.3	(l) 33.2	(m) 49.4	(n) 9.7	(o) 80	(p) 0.629

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.6	80.6	68.4	76.6	66.4	73.5	64.7	69.8	78.1	68.0	74.7	66.2	71.8	9.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.9	99.0	73.9	65.3	93.5	71.6	63.8	88.6	69.5	33.8	78.2	32.3	74.8	30.9	72.1	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB
(a) 29.3	(b) 25.4	(c) 22.3	(d) 16.2	(e) 87.9	(f) 8.9	(g) 3.2	(h) 9.8	(i) 90.2	(j) 4.6	(k) 92.1	(l) -0.4	(m) 93.9	(n) -6.8	(o) 96.2	(p) 78.7	
(a) 29.3	(b) 25.4	(c) 22.3	(d) 15.7	(e) 74.2	(f) 8.6	(g) 1.7	(h) 9.5	(i) 75.4	(j) 4.4	(k) 76.4	(l) -0.4	(m) 77.4	(n) -6.6	(o) 78.7	(p) 78.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	37.9	38.7	42.5	48.7	54.5	59.8	63.6	63.0	58.4	52.0	45.2	39.9
	DBStd	10.86	7.21	6.78	6.14	6.34	6.04	5.04	5.02	4.36	4.69	5.62	5.62	6.97
	HDD50	1574	375	317	240	100	21	0	0	0	0	45	161	315
	HDD65	5424	839	736	698	490	330	171	87	92	206	405	593	777
	CDD50	1729	1	1	7	60	161	294	423	404	252	105	18	3
	CDD65	103	0	0	0	0	6	15	45	30	7	0	0	0
	CDH74	636	0	0	0	4	33	116	308	148	27	0	0	0
	CDH80	134	0	0	0	0	2	24	79	26	3	0	0	0
	WSAvg	10.8	13.1	12.6	12.2	10.4	10.6	9.8	9.5	9.1	9.2	10.6	10.9	12.2
	PrecAvg	35.9	3.0	2.4	2.0	1.7	2.2	2.5	3.1	4.0	3.6	3.9	3.9	3.4
(15) Precipitation	PrecMax	48.4	6.1	5.0	4.8	4.1	4.1	6.3	6.5	9.3	7.8	7.8	6.9	6.5
	PrecMin	26.0	0.3	0.7	0.4	0.0	0.9	0.9	0.7	0.9	0.4	0.9	0.4	0.9
	PrecStd	5.4	1.4	1.1	1.2	1.0	0.8	1.3	1.5	2.2	1.8	1.8	1.5	1.4
	PrecStd	5.4	1.4	1.1	1.2	1.0	0.8	1.3	1.5	2.2	1.8	1.8	1.5	1.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	53.6	56.5	62.5	73.2	77.9	83.5	86.9	83.6	77.6	69.5	60.2	55.7	53.7
	0.4% MCWB	51.3	50.7	53.4	59.5	64.5	69.5	70.4	70.6	67.6	62.5	57.7	53.7	53.7
	2% DB	51.2	51.4	57.3	68.1	73.5	77.1	81.4	78.1	73.0	64.7	57.4	53.1	53.1
	2% MCWB	49.3	48.6	51.1	57.8	61.9	66.5	68.9	68.1	65.6	60.2	55.2	51.2	51.2
	5% DB	49.2	49.4	53.6	63.3	69.4	72.7	77.0	74.1	69.5	62.1	55.3	51.1	51.1
	5% MCWB	47.4	47.0	49.0	55.0	60.1	64.5	66.9	65.8	63.3	58.6	53.1	49.1	49.1
	10% DB	47.4	47.5	51.0	59.2	65.4	69.2	73.3	71.1	66.5	60.1	53.3	48.9	48.9
	10% MCWB	45.7	45.4	47.6	52.8	58.2	62.4	65.1	64.2	61.5	57.2	51.0	47.1	47.1
	0.4% WB	51.6	51.3	54.7	61.6	67.1	71.0	72.6	72.0	69.1	63.3	57.9	53.8	53.8
	0.4% MCDB	53.3	53.8	60.3	70.8	74.6	79.7	83.1	81.2	75.3	68.2	59.4	55.0	55.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	49.6	49.3	51.8	58.3	63.6	68.0	70.2	69.0	66.3	60.8	55.5	51.7	51.7
	2% MCDB	50.9	51.1	55.8	66.3	70.7	74.8	79.2	76.0	71.8	64.0	57.2	53.0	53.0
	5% WB	47.7	47.3	49.7	55.8	60.9	65.4	68.3	67.0	64.2	59.2	53.4	49.3	49.3
	5% MCDB	48.9	49.1	53.1	62.3	67.9	71.5	75.7	72.5	68.3	61.7	55.1	50.7	50.7
	10% WB	45.9	45.6	47.9	53.5	58.6	63.2	66.2	65.3	62.4	57.5	51.4	47.4	47.4
	10% MCDB	47.4	47.4	50.6	58.5	64.7	68.3	71.8	69.9	65.9	59.7	53.0	48.8	48.8
(35) Mean Daily Temperature Range	MDBR	7.7	9.1	12.2	16.0	15.3	15.0	14.6	14.2	14.0	11.5	8.9	7.8	7.8
	5% DB	7.7	10.6	16.3	22.0	21.3	20.7	21.0	18.9	17.4	12.8	9.0	7.4	7.4
	5% WB	7.5	8.5	11.2	13.0	11.6	11.8	10.4	10.1	10.6	9.3	7.7	7.1	7.1
	MCDBR	7.7	9.5	15.2	20.6	19.3	18.4	19.3	16.7	15.7	12.4	8.7	7.4	7.4
	MCWBR	7.7	8.2	11.0	12.9	11.1	11.5	10.5	9.7	10.1	9.5	7.7	7.3	7.3
	MCWBR	7.7	8.2	11.0	12.9	11.1	11.5	10.5	9.7	10.1	9.5	7.7	7.3	7.3
(40) Clear-Sky Solar Irradiance	taub	0.341	0.360	0.391	0.412	0.414	0.411	0.422	0.414	0.394	0.374	0.352	0.334	0.334
	taud	2.301	2.303	2.237	2.232	2.285	2.311	2.290	2.328	2.373	2.367	2.323	2.295	2.295
	Ebn at Noon	199	233	249	260	266	267	262	257	247	226	193	178	178
	Edh at Noon	20	27	35	40	39	39	39	36	30	25	20	17	17
	RadAvg	210	425	826	1341	1647	1739	1660	1380	981	552	243	156	156
(45) All-Sky Solar Radiation	RadStd	20	75	87	125	99	109	166	107	82	49	22	17	17

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (67 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page